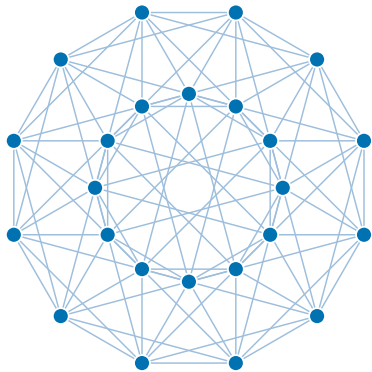
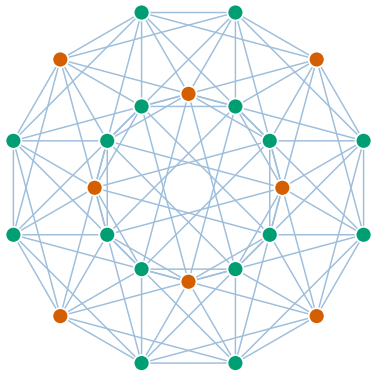


(a) D_4 : $(\pm 1, \pm 1, 0, 0)$



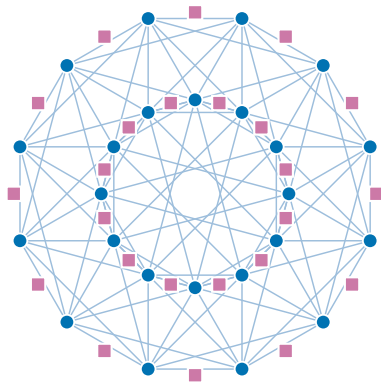
24 vectors, $|e|^2 = 2$
one symmetry orbit

(b) D_4^* : $(\pm 1, 0, 0, 0) \oplus (\pm \frac{1}{2}, \dots)$



8 axial + 16 diagonal, $|e|^2 = 1$
still one orbit: triality mixes them

(c) shell 1 and shell 2 of D_4



two interlocking 24-cells
the self-duality $D_4^* \simeq D_4$